

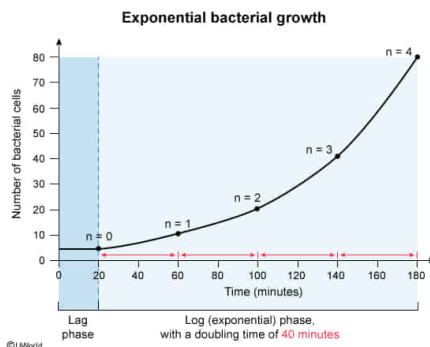
A student adds 5 bacterial cells to a test tube containing fresh medium and incubates it for 3 hours at 37°C. The bacterial population shows an initial lag phase of 20 minutes followed thereafter by a growth phase with a doubling time of 40 minutes. Given this, what is the approximate number of bacterial cells present at the end of the incubation period? (Note: Assume unrestricted growth during incubation.)

- ☐ A. 20 cells [7%]
☐ B. 40 cells [18%]
☒ C. 80 cells [67%]
☐ D. 160 cells [7%]

Correct

66% answered correctly

Explanation:



Binary fission is the process by which unicellular organisms such as bacteria reproduce asexually. During this process, the **parental bacterial cell** doubles in size and then **divides** into **two identical daughter cells**. Subsequently, each daughter cell also divides into two, yielding four cells that then continue to divide in a similar manner. This type of growth is called exponential (ie, growth in which the rate of reproduction increases over time).

The **generation time** (doubling time) is the time needed for a bacterial population to double, or the time needed for a single bacterium to divide into two. When a single cell divides by binary fission, the number of cells that result after a specified time is calculated by multiplying the original number of bacterial cells by 2^n , where n equals the number of generations.

The rate at which a bacterial population grows can be demonstrated by plotting a typical **growth curve**, which has a lag, log, stationary, and death phase. Before they begin dividing, transplanted bacteria often exhibit an initial lag phase as they become accustomed to the local conditions and synthesize the components (eg, enzymes, RNA) required for metabolism. In the subsequent log phase, the bacterial population grows exponentially as bacteria are able to effectively metabolize nutrients.

According to the question, 5 bacterial cells were incubated for 3 hours (180 minutes); these exhibited a 20-minute lag phase and a 40-minute generation time. Assuming unrestricted growth during incubation, the bacterium could only have had a log phase and a lag phase. Therefore, the number of cells that result after incubation can be calculated

as follows:

1. Calculate the amount of time in which the bacterial population was in log phase:
 $180 \text{ min incubation time} - 20 \text{ min lag phase} = 160 \text{ min log phase}$
2. Find the number of generations in the log phase period:
 $160 \text{ min log phase} / 40 \text{ min generation time} = 4 \text{ generations}$
3. Multiply the original number of bacterial cells by 2^n to get the final population size:
 $5 \text{ bacterial cells} \times 2^4 = \mathbf{80 \text{ cells}}$

Educational objective:

Binary fission is the process by which a unicellular organism (eg, bacterium) divides into two identical daughter cells. The generation time refers to the time in which a bacterial population doubles. When a single cell divides by binary

fission, the number of cells that result is calculated by multiplying the original number of bacterial cells by 2^n , where n equals the number of generations.

Time spent:0 sec
Subject:
Biology

QID:401292
System:
Cellular Biology

Assume a single-celled organism was discovered in water above 100°C. On closer inspection, it is determined that this organism is classified under the Archaea domain. Given this information, this organism most likely has:

- ☐ A. a peptidoglycan layer in its cell wall. [24%]
- ✓ ☒ B. a circular chromosome. [62%]
- ☐ C. telomeres. [4%]
- ☐ D. an endoplasmic reticulum. [8%]

Correct

62% answered correctly

Explanation:

Characteristics	Bacteria	Archaea	Eukarya
Cell type	Unicellular	Unicellular	Unicellular or multicellular
Nucleus	Absent	Absent	Present
Organelles	Absent	Absent	Present
Cell wall with peptidoglycan	Present	Absent	Absent
Method of cellular division	Binary fission	Binary fission	Mitosis
Chromosome	Circular	Circular	Linear

Cells can exist in many forms and sizes; some live as independent (free-living) organisms, and others live as a group in a colony or make up the body of a multicellular organism. Based on their shared characteristics, all living organisms can be classified into one of **three domains**: Bacteria, Archaea, or Eukarya.

Eukaryotic organisms are classified into the Eukarya domain, but **prokaryotic** organisms are in either the Bacteria or the Archaea domain. In this scenario, the organism is in the Archaea domain. Organisms classified as **archaea share traits** with **bacteria**. Like bacteria, archaea are **unicellular** (single celled), have a **circular chromosome**, have **no membrane-bound organelles** or nucleus, and reproduce asexually via **binary fission**. However, unlike bacteria, archaea *do not* have the chemical **peptidoglycan** in their cell walls (**Choice A**).

(**Choice C**) **Telomeres** are regions of repetitive nucleotide sequences at the ends of eukaryotic chromosomes that **shorten** with each round of cell division. Telomeres are present only on linear *eukaryotic* chromosomes and protect these ends from deteriorating. In contrast, both archaea and bacteria have circular chromosomes (*lacking* telomeres).

(**Choice D**) The **endoplasmic reticulum**, along with other membrane-bound organelles (eg, **Golgi body**, nucleus), is found only in eukaryotic cells. Archaea and bacteria are known as prokaryotes because they lack membrane-bound organelles and have no nucleus.

Educational objective:

Prokaryotes are unicellular organisms that can be further classified into the Archaea domain or the Bacteria domain. Organisms in these domains lack a nucleus and membrane-bound organelles but have circular chromosomes and are able to reproduce asexually through binary fission. However, bacteria (not archaea) have peptidoglycan in their cell walls as a distinguishing feature.

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